

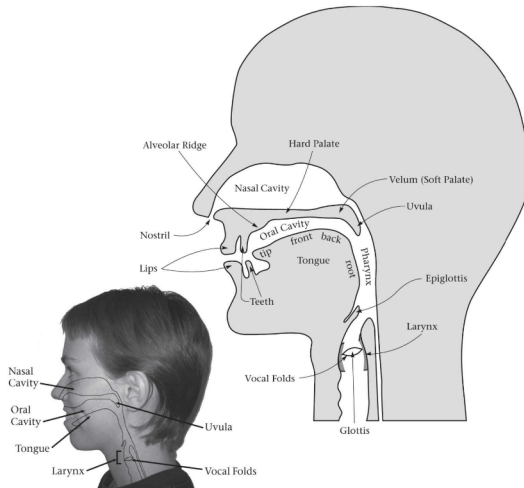
Problem Session 1

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LING 101.02 - Fall 2026

September 30, 2026

Vocal Tract



Consonants: Stops

Caution: Some sounds may contain the risk of suffocation during pronunciation. Do not try them without the presence of a phonetician.

Let's start reviewing stop consonants first. Stops, also known as plosives, are speech sounds produced by obstructing airflow in the vocal tract, creating a build-up of pressure that is released suddenly.

[p] pit, tip, spit, hiccough, appear

[b] ball, globe, amble, brick, bubble

[t] tag, pat, stick, pterodactyl, stuffed

[d] dip, card, drop, loved, batted

[k] kit, scoot, character, critique, exceed

[g] guard, bag, finger, designate, Pittsburgh

[ʔ] uh-oh, hatrack, Batman, button, curtain

Consonants: Fricatives

Fricatives are speech sounds produced by forcing air through a narrow constriction in the vocal tract, creating turbulence.

[f]	foot, laugh, philosophy, coffee, carafe
-----	---

[v]	vest, dove, gravel, anvil, average
-----	------------------------------------

[θ]	through, wrath, thistle, ether, teeth
-----	---------------------------------------

[ð]	the, their, mother, either, teethe
-----	------------------------------------

[s]	soap, psychology, packs, descent, peace
-----	---

[z]	zip, roads, kisses, Xerox, design
-----	-----------------------------------

[ʃ]	shy, mission, nation, glacial, sure
-----	-------------------------------------

[ʒ]	measure, vision, azure, casualty, decision or, leisure
-----	---



Consonants: Affricates

Affricates are speech sounds that begin as stops (complete closure of the vocal tract) and release as fricatives (partial closure that creates turbulence).

[tʃ] choke, match, feature, constituent

[dʒ] judge, George, Jell-O, region, residual

Consonants: Approximants

Approximants are speech sounds produced with a relatively open vocal tract, allowing for airflow without creating turbulent noise.

[l]	leaf, feel, Lloyd, mild, applaud
-----	----------------------------------

[ɹ]	reef, fear, Harris, prune, carp
-----	---------------------------------

[w]	with, swim, mowing, queen, twilight
-----	-------------------------------------

[j]	you, beautiful, feud, use, yell
-----	---------------------------------

Consonants: Nasal

The velum (the soft part of the roof of the mouth) lowers to allow airflow through the nose, creating these distinct sounds.

[m]	moose, lamb, smack, amnesty, ample
-----	------------------------------------

[n]	nap, design, snow, know, mnemonic
-----	-----------------------------------

[ŋ]	lung, think, finger, singer, ankle
-----	------------------------------------

Fill in the table

| Some Phonetic Symbols for American English Consonants

Bilabial

Labiodental

Interdental

Alveolar

Palatal

Velar

Glottal

Stop (oral)

voiceless

voiced

Nasal (voiced)

Fricative

voiceless

voiced

Affricate

voiceless

voiced

Glide

voiceless

voiced

Liquid (voiced)

(central)

(lateral)

Fill in the table

| Some Phonetic Symbols for American English Consonants

	Bilabial	Labiodental	Interdental	Alveolar	Palatal	Velar	Glottal
Stop (oral)							
voiceless	p			t		k	ʔ
voiced	b			d		g	
Nasal (voiced)						-	
Fricative							
voiceless							
voiced							
Affricate							
voiceless							
voiced							
Glide							
voiceless							
voiced							
Liquid (voiced)							
(central)							
(lateral)							

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Nasal (voiced)	m			n		ŋ	
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voiceless							
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Affricate							
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Glide							
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Fricative							
voiceless		f	θ	s	ʃ		h
voiced		v	ð	z	ʒ		
Affricate							
voiceless							
voiced							
Glide							
voiceless							
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Liquid (voiced)							
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Affricate							
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Glide							
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Affricate							
voiceless					tʃ		
voiced					dʒ		
Glide							
voiceless	ɱ					ɰ	
voiced	w				j	w	
Liquid (voiced)							
(central)							
(lateral)							

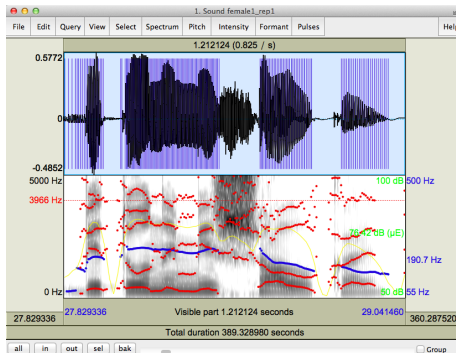
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voiced					dʒ		
Glide							
voiceless	ɱ					ɰ	
voiced	w				j	w	
Liquid (voiced)							
(central)				r			
(lateral)				l			

Showing Praat

Now I will show you Praat, which is a speech analysis tool used for doing phonetics by computer. We will analyze some of these consonants together and investigate their phonetic properties better.



Vowels

[i] beat, we, believe, people, money, dean

[ɪ] bit, consist, injury, malignant, business, gym

[e] bait, reign, great, they, gauge, age

[ɛ] bet, reception, says, guest, bend

[æ] bat, laugh, anger, comrade, rally, hand

[u] boot, who, brewer, duty, through, dune

[ʊ] put, foot, butcher, could, boogie-woogie

[o] oh, boat, beau, grow, though, over

[ɑ] pot, father, sergeant, honor, hospital, bomb

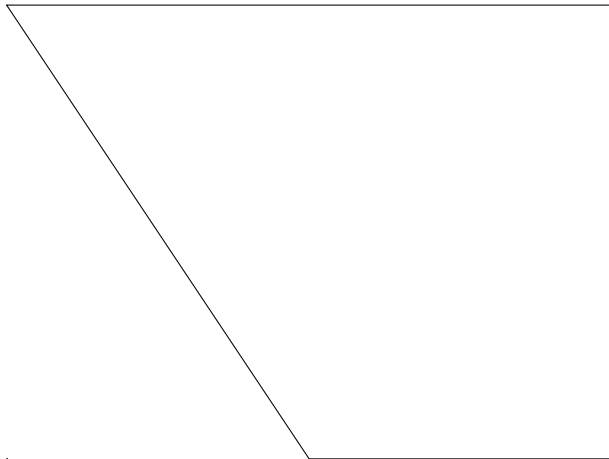
[ʌ] but, tough, another, oven, fungus

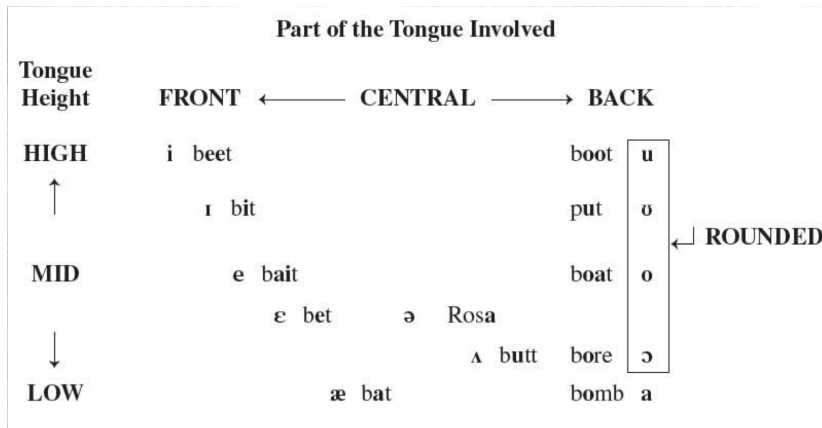
[ə] among, Asia, eloquent, famous, harmony

Vowels: Practice Questions

- What are the four qualities we use to distinguish vowels?
- We also have vowels in Turkish that are not present in English. Name them and try to describe their properties.
- Check out Jack The Linguist's Song: Vowel Space: [here](#)

Vowel Chart: Exercise





Let's work on some practice questions to test our understanding of phonetic symbols and concepts.

Write the Phonetic Symbols

Give the phonetic symbols for the following segments:

- voiceless velar stop
- voiced alveolar fricative
- alveolar nasal
- voiceless alveolar stop
- voiceless labiodental fricative
- voiced interdental fricative
- voiceless glottal fricative

Phonetic symbols for the following segments:

- voiceless velar stop: [k]
- voiced alveolar fricative: [z]
- alveolar nasal: [n]
- voiceless alveolar stop: [t]
- voiceless labiodental fricative: [f]
- voiced interdental fricative: [ð]
- voiceless glottal fricative: [h]

[ʃʊt]	shut
-------	------

[swɪt]	swift
--------	-------

[fəʊ]	follow
-------	--------

[trəd]	trod
--------	------

[lɛft]	left
--------	------

[tʃɪld]	child
---------	-------

[ðɪm]	theme
-------	-------

[vɔɪs]	voice
--------	-------

[ræŋ]	rang
-------	------

[hɛlθ]	health
--------	--------

- [ʃʌt]
- [swɪft]
- [fəloʊ]
- [tʃaɪld]
- [θɪm]
- [voɪs]
- [ræŋ]

Important Resource for Phonetics

There is a cool website "A Course in Phonetics" developed by UC Berkeley Linguistics, which offers resources for phonetics:

<https://linguistics.berkeley.edu/acip/>

Also check out the following websites:

- <http://home.cc.umanitoba.ca/~krus-sll/phonetics/practice/consonant/drill2.htm>
- <https://www.phon.ucl.ac.uk/cgi-bin/wtutor?tutorial=siphtra/plostut1/plostut1.htm>
- <https://www.seeingspeech.ac.uk/ipa-charts/?chart=1datatype=3speaker=1>
- [https://users.castle.unc.edu/~jlsmith/phonetics-resources.html\(0\)](https://users.castle.unc.edu/~jlsmith/phonetics-resources.html(0))
- <http://tophonetics.com>